HAKIM TRACHET

Master's research student in Automated Manufacturing Engineering, École de technologie supérieure (ÉTS)
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PROFILE

Research student in machine learning applied to aircraft piloting. I compare deep sequence architectures that predict a pilot's commands during takeoff, then fly them in closed loop in a simulator. General engineering training (Arts et Métiers) in a dual degree with ÉTS.

EDUCATION

2025 – present **Master's degree (thesis-based) in Automated Manufacturing Engineering**, ÉTS, Montréal
Cumulative GPA: 4.18/4.30 (15 course credits completed), with A+ in Deep Learning, Neural Networks and Fuzzy Systems, Data Analysis and Predictive Systems, and Directed Readings.
Dual degree with Arts et Métiers. Supervisor: Georges Ghazi (ÉTS). Co-supervisor: Ange Adrienne Nyamen Tato (Université Laval, member of the Institute Intelligence and Data, IID).
Thesis: *Optimization of an ontology-based pilot simulation model to reproduce human behaviour*.

2023 – 2025 **Grande École Engineering Program (diplôme d'ingénieur)**, Arts et Métiers (ENSAM), Aix-en-Provence
General engineering degree, mechanical and energy engineering. 124 ECTS credits earned (L3 and M1 levels), including ten course units graded A (top 10 %).

2021 – 2023 **Classes préparatoires aux grandes écoles (PTSI-PT)**, Lycée Jean Mermoz, Montpellier
Two-year intensive program in mathematics, physics and engineering sciences that precedes admission to the grandes écoles. Class council commendation in both PTSI semesters.

2021 **French Baccalauréat, highest honours (mention Très bien)**, Lycée Jean Monnet, Montpellier

RESEARCH EXPERIENCE

Since 2025 **Graduate Research Assistant, C-Pilot project**, ÉTS and Université Laval
Project supported by CRIAQ and NSERC, with Bombardier, Beam Me Up Augmented Intelligence and Cognitive Group as industry partners. Work packages: learned models and simulator deployment.

- Compared four deep sequence architectures (TCN-LSTM, Transformer, MLP-Mixer, TS-Mixer) across three takeoff phases and five window sizes (60 configurations), with Bayesian hyperparameter optimization (Optuna) and five-fold cross-validation.
- Selected models with multi-criteria methods (WPM and TOPSIS) and tested the robustness of the choice with a sensitivity analysis over 1,771 weight sets; retained the MLP-Mixer, with R^2 of 0.946 at rotation, 0.909 in initial climb and 0.662 in the ground roll, at no more than 1.75 ms of compute per step.
- Exported the models to ONNX with built-in normalization and deployed them in Simulink and Python; achieved complete, repeatable closed-loop takeoffs at 50 Hz in X-Plane 12.
- Refactored the research code into an installable, tested Python package (36 modules, 154 tests) for transfer to industry partners; contributed to project deliverables.

ENGINEERING PROJECT

XPlaneCom: from learned model to closed-loop synthetic pilot

A model that predicts well offline does not necessarily fly well: once connected to the aircraft, its errors propagate from one step to the next. I therefore designed XPlaneCom, a platform that flies the learned models in X-Plane 12 at 50 Hz and checks every trial.

- Prepare the flight (mass, centre of gravity, position on the runway), run the takeoff and log every step.
- Replay the model on the log to confirm that the command sent is the one computed, and compare trials with one another in a web interface.
- Provide an assistance mode that blends the model's command with that of a flight director corrector, with a tracking measure.
- **Result**: complete, repeatable takeoffs, with a median inference time of 0.76 ms, and evidence of a slower rotation than the pilots', which no offline metric showed.
- **Tools**: Python, ONNX Runtime, Simulink, X-Plane 12, web interface; 118 automated tests.

PUBLICATIONS AND PRESENTATIONS

Trachet, H., Neffati, C., Ghazi, G., & Nyamen Tato, A. A. (2027). *Deep sequence models for pilot command prediction during takeoff: A multi-criteria comparative evaluation of TCN-LSTM, Transformer, MLP-Mixer, and TS-Mixer*. Conference paper, AIAA SciTech Forum 2027, Orlando, Florida, January 11–15, 2027.

Trachet, H., Neffati, C., Ghazi, G., & Nyamen Tato, A. A. (2026). *From simulation to deployment: Deep learning models for pilot command prediction in real-time takeoff scenarios*. Oral presentation, CASI 2026 conference, Winnipeg, Manitoba, October 14–16, 2026.

Neffati, C., **Trachet, H.**, Ghazi, G., & Nyamen Tato, A. A. (2026). Temporal clustering of flight simulator data for learner profiling: A methodological comparison. Poster, *Proceedings of the 19th International Conference on Educational Data Mining (EDM 2026)*, Seoul, 780–785. doi:10.5281/zenodo.21040146

EXPERIENCE AND INVOLVEMENT

Sept. – Dec. 2024 — **Lead, integration of the incoming class**, Arts et Métiers, Aix-en-Provence
Led a team of 12: set a daily schedule, assigned tasks, held daily check-in meetings and wrote the reports.

June – July 2024 — **Warehouse intern, parcel handling**, high-volume logistics warehouse
Sorted parcels by size and delivery zone, updated tracking data and coordinated with drivers and the logistics team.

Summers 2023–24 — **Volunteer event decorator**
Designed layouts for indoor and outdoor event spaces; chose materials, colours and textures to create an atmosphere consistent with a theme.

PROJECTS

2024 — **System and Product Design project**, Arts et Métiers
Built 3D models and assemblies, verified structural integrity and analyzed numerical simulation results.

2023 — **Rubik's Cube simulator**, supervised personal project
Developed a Python simulator with an interactive interface for 3D visualization and manipulation in FreeCAD.

2022 – 2023 — **TIPE, automated parallel parking**, supervised personal project, preparatory classes
Designed an algorithm adapted to the dimensions of the parking space and validated it on a miniature car programmed with Arduino.

SKILLS

Machine learning	PyTorch, Optuna, scikit-learn, ONNX and ONNX Runtime; time series, imitation learning, multi-criteria selection
Programming	Python (NumPy, pandas, SciPy, Matplotlib), automated testing (pytest), LaTeX
Simulation and CAD	Simulink, X-Plane 12, 3DExperience, Abaqus, FreeCAD, Arduino, Excel
Languages	French (native), Arabic (fluent), English (B2, scientific communication in English), Spanish (basic)
Interests	Club football, hiking and exploring landscapes